Two-coordinate affine rotation table: report

Question and scope

Does storing an additional transformed x coordinate make radix-14 affine-layer batch evaluation faster by replacing constant field multiplications during leaf selection? The candidate stores `(x, m, y)` with `m=x+βx`. The identities `m−x=βx` and `−m=β²x` follow from `β²+β+1=0`; TwoX.lean checks the abstract additive-group implications with explicit hypotheses. The native overlay is hash-guarded, and independent binary replay checks the curve-specific implementation.

Result

The frozen CPU 40 panel used 16 alternating repetitions for the comparison plus adjacent A/A controls. All 96 arms and 48 pairs completed, source/binary/fixture artifacts were stable, every output was verified, and all A/A ratios passed `[0.90, 1.11]`. The independent audit is `tier1-two-x-cpu40-audit.json`; it checks every arm against the manifest, fixture and hashes.

Metric	Affine-layer reference	Two-x candidate
Mean online time, 12,288 outputs	56.458543 ms	54.416974 ms
Mean time per scalar	4.594608 µs	4.428465 µs
Mean table preparation	4,196.115 ms	4,230.430 ms
Retained bytes	26,035,988	37,192,948

The predeclared median paired reference/candidate ratio is **1.037997×**, with paired bootstrap 95% interval **[1.025659, 1.053525]** over 16 comparison repetitions. This qualifies a batch-throughput comparison under the frozen panel on the observed CPU 40 instance. It does not establish a one-scalar latency gain; physical-host exclusivity is unverified. The candidate uses 11.16 MiB more retained table memory and table preparation is 34.315 ms longer in the means.

The generic runner's `paired_speedup=1.006040` includes the two A/A case medians and is not the comparison estimate. The case-specific audit applies A/A only as noise gates. The overlay's source hashes, native release binary, and retained first failed compile are described in the README and provenance archive.

The provenance archive contains the post-overlay Rust source snapshot and Cargo lock, but not the pinned input sources (`f3c3b072…` and `cecb27d0…`) or their originating repository/revision. A local build from the archive stops before compilation because the source references 33 external absolute-path include files that are not included. The captured hashes, build/test logs, and receipt audit remain available, but the native overlay and build are not independently reproducible from this PR alone.

The 96 run records contain a cgroup memory high-water value of 2,184,867,840 bytes, shared by the container, but no per-process peak RSS. The general hardware report is incomplete at process granularity; this is separate from the explicit Tier 1 timing criteria, which the panel passes.

Protocol and figures

The panel manifest and rules freeze the 12,288-output fixture, CPU affinity, case order, repetition count and A/A range. Online time excludes table preparation, which is measured separately; every result is verified after timing. The report figures are generated from the committed audit JSON by `visuals.py`, with their SVG and PDF sources committed here. Regenerate the figures from the audit JSON and rebuild the PDF from the committed SVGs with:

```
python3 experiments/prime-j0-affine-two-x-20261015/visuals.py
pandoc experiments/prime-j0-affine-two-x-20261015/REPORT-2026-10-19.md \
  --from=gfm --pdf-engine=typst -V papersize=us-letter \
  --resource-path=experiments/prime-j0-affine-two-x-20261015 \
  -o experiments/prime-j0-affine-two-x-20261015/REPORT-2026-10-19.pdf
```

Representation diagram (SVG) · PDF



